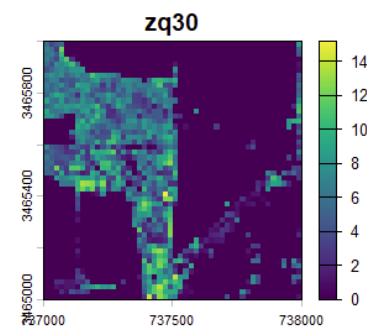
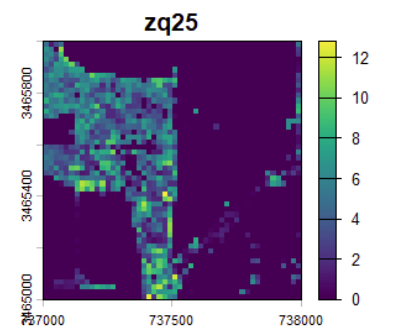
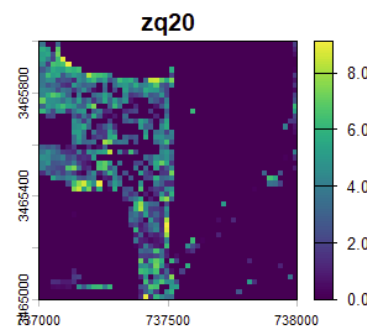
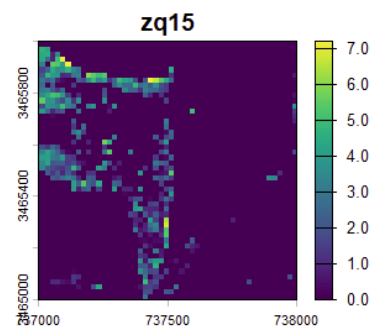
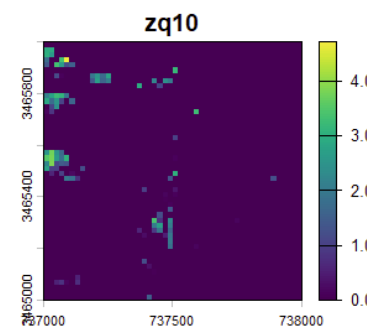
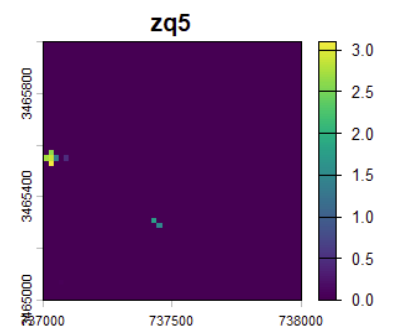
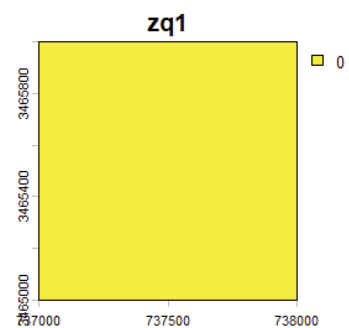
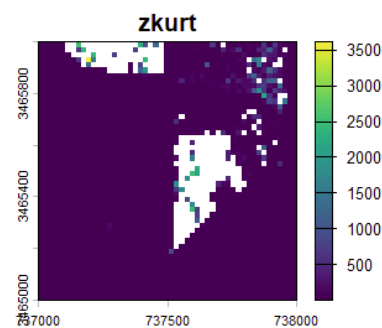
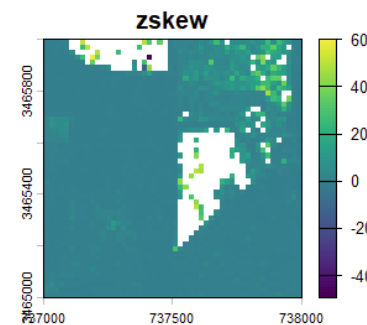
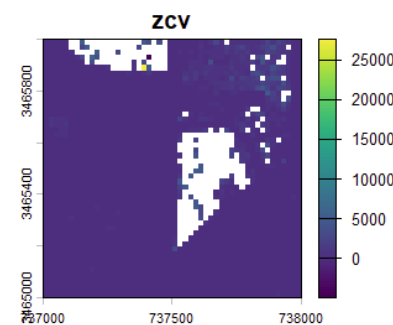
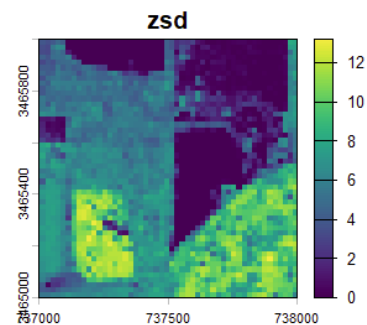
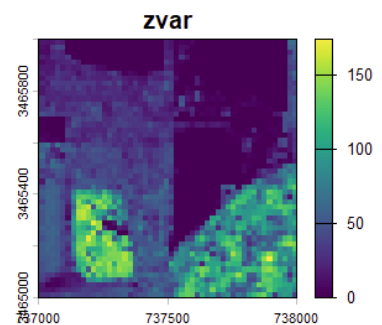
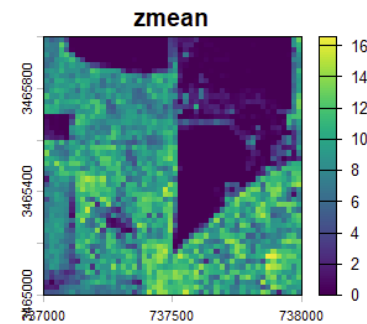
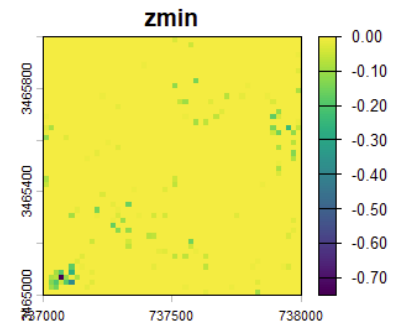
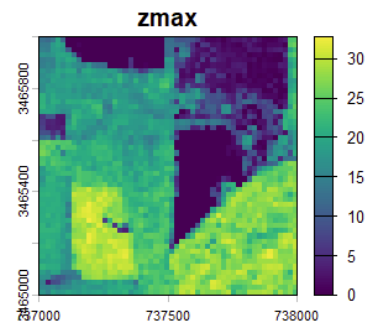
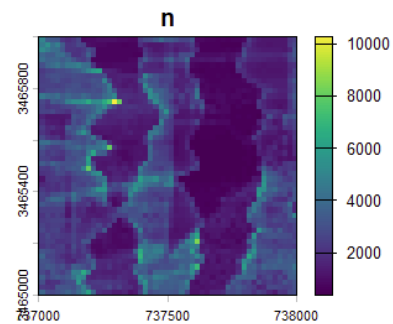
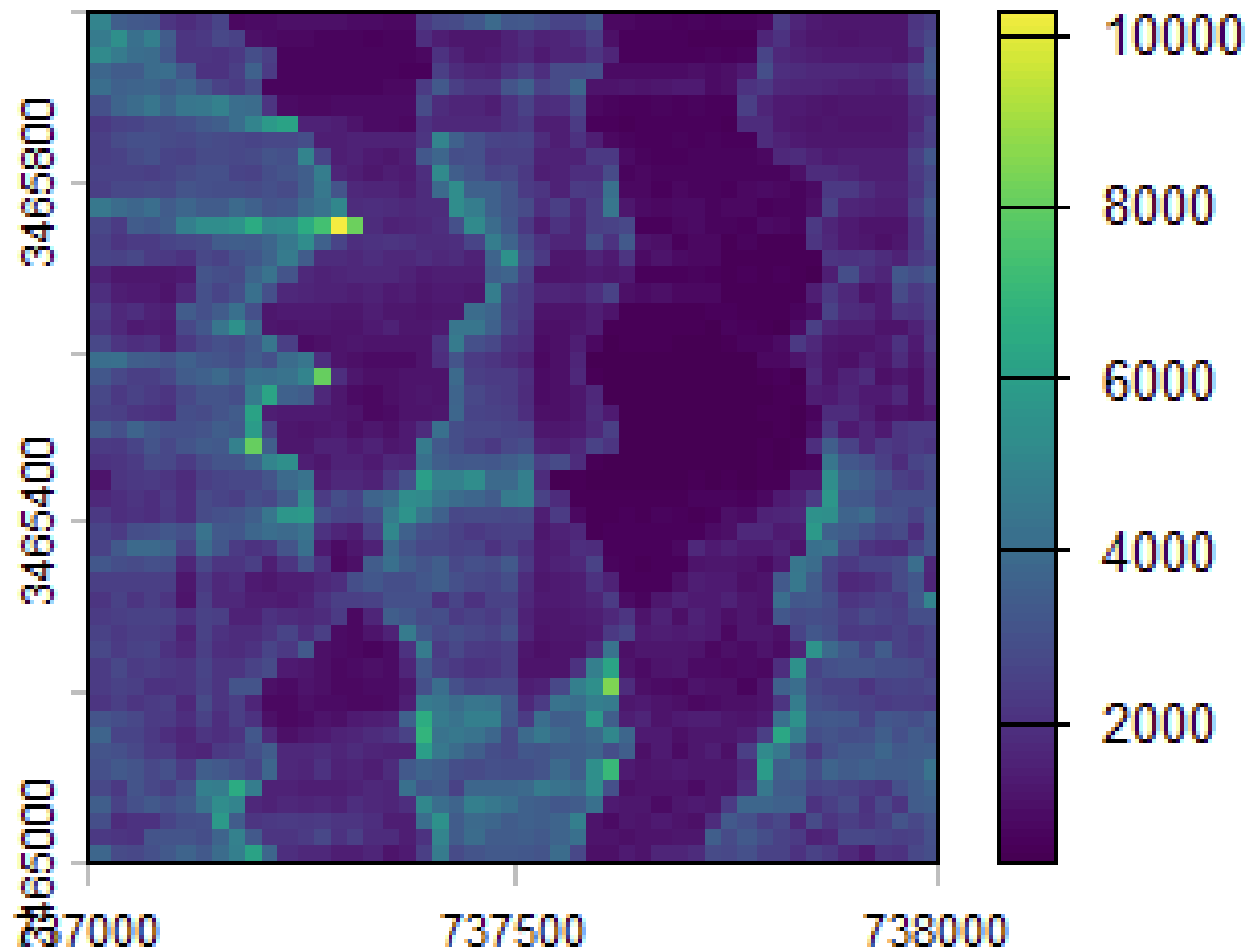


Reduce density but preserve structural information

```
Master_metrics_testing_tile.R x Master_metrics_function.R x
Source on Save
1 library(lidR)
2 library(terra)
3 library(lidRmetrics)
4 library(Lmoments)
5 install.packages("geometry")
6
7 source("Master_metrics_function.R")
8
9 las <- readLAS("NEON_lidar_tile.laz", filter = "-keep_random_fraction 0.5")
10
11 las <- readLAS("NEON_lidar_tile.laz", filter = "-thin_with_voxel 0.25")
12
13 table(las$Classification)
14 # Appears to be some buildings labeled 6 -> LASBUILDING
15 las <- filter_poi(las, !Classification %in% c(LASNOISE, LASBUILDING))
16 las <- normalize_height(las, algorithm = tin())
17
18 metric_stack <- pixel_metrics(las, ~master_metrics(X, Y, Z, Intensity, ReturnNumber, NumberOfReturns,
19                                     res = 20)
20 names(metric_stack)
21 #108 different layers
22 plot(metric_stack)
```



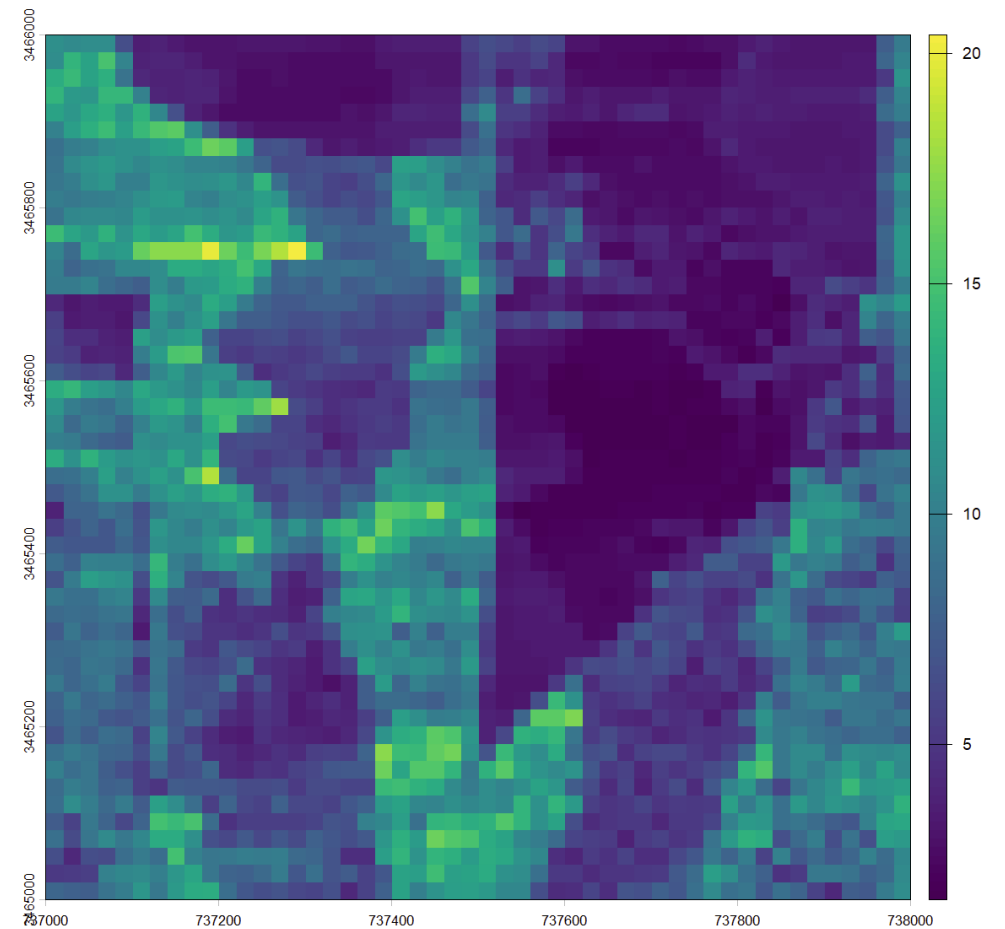
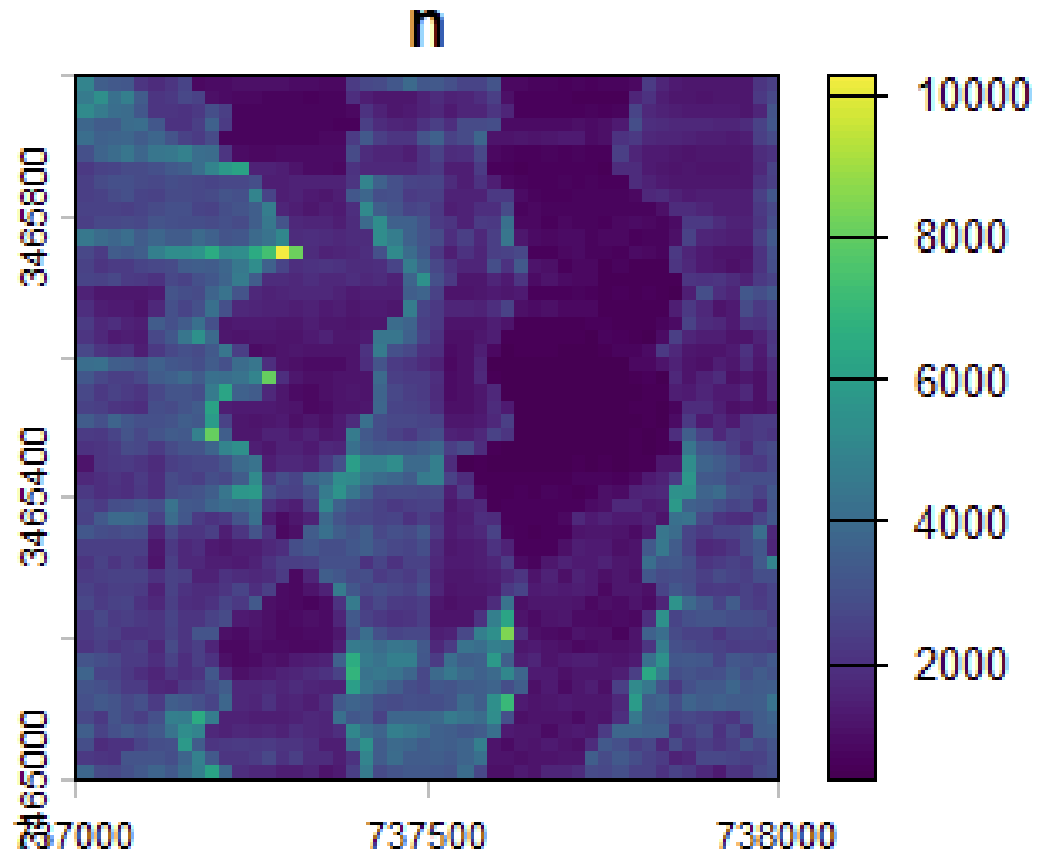
n



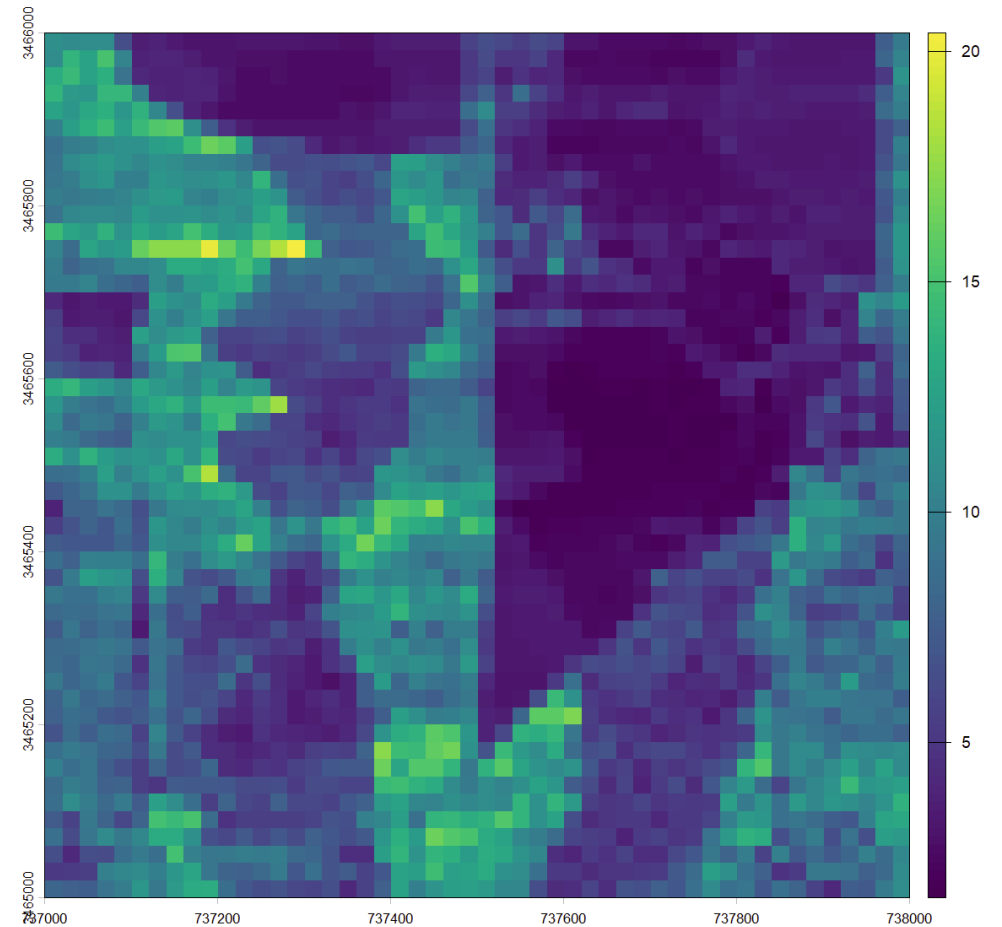
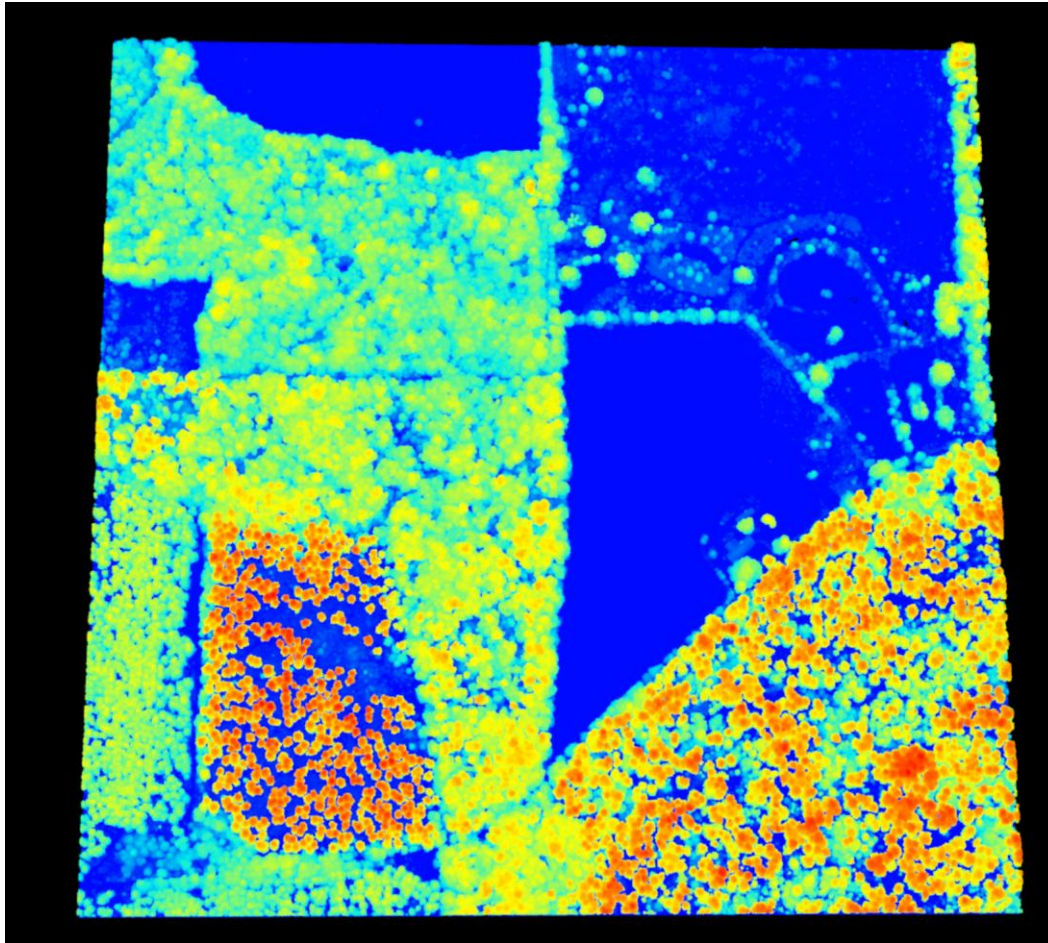
```

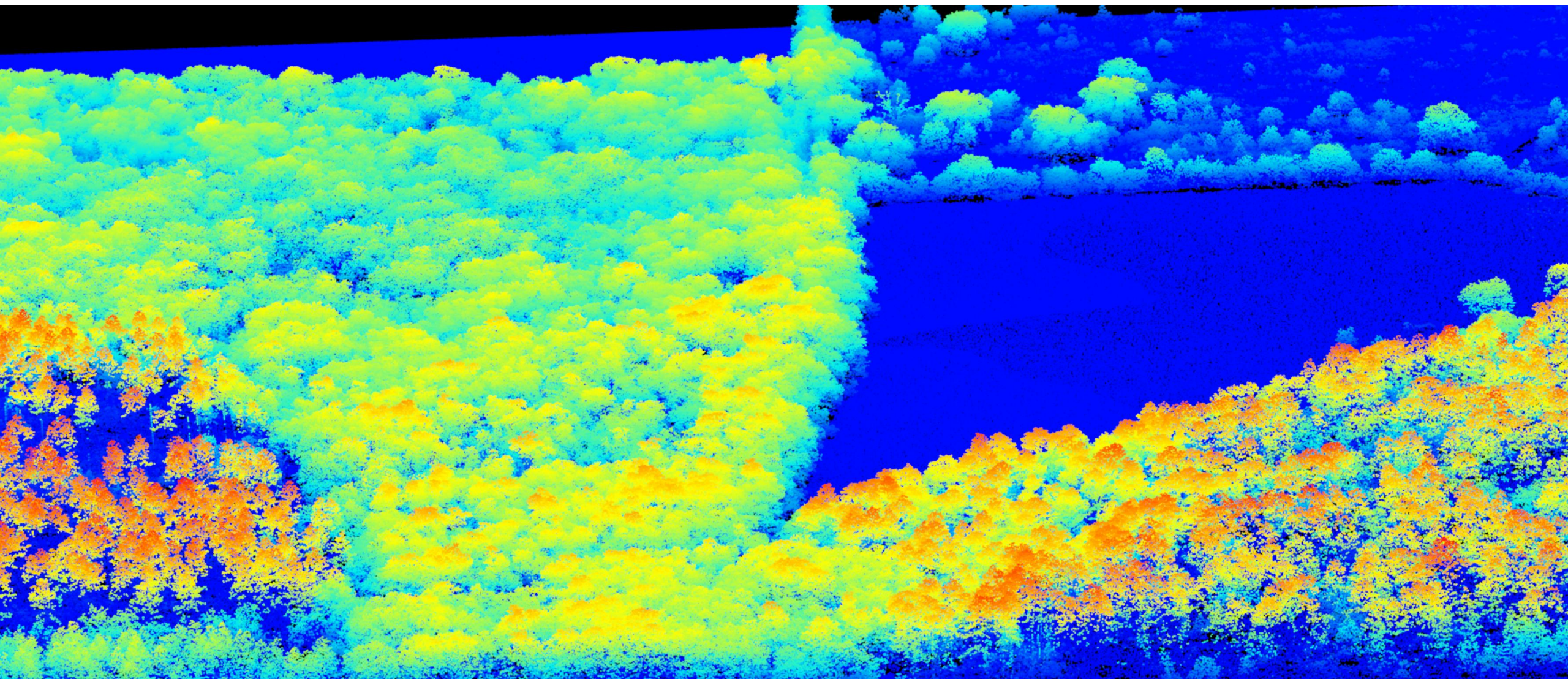
8
9 #las <- readLAS("NEON_lidar_tile.laz", filter = "-keep_random_fraction 0.5")
10 las <- readLAS("NEON_lidar_tile.laz", filter = "-thin_with_voxel 0.5")
11
12 x = pixel_metrics(las, ~length(Z)/(20*20), res=20)
13 plot(x)

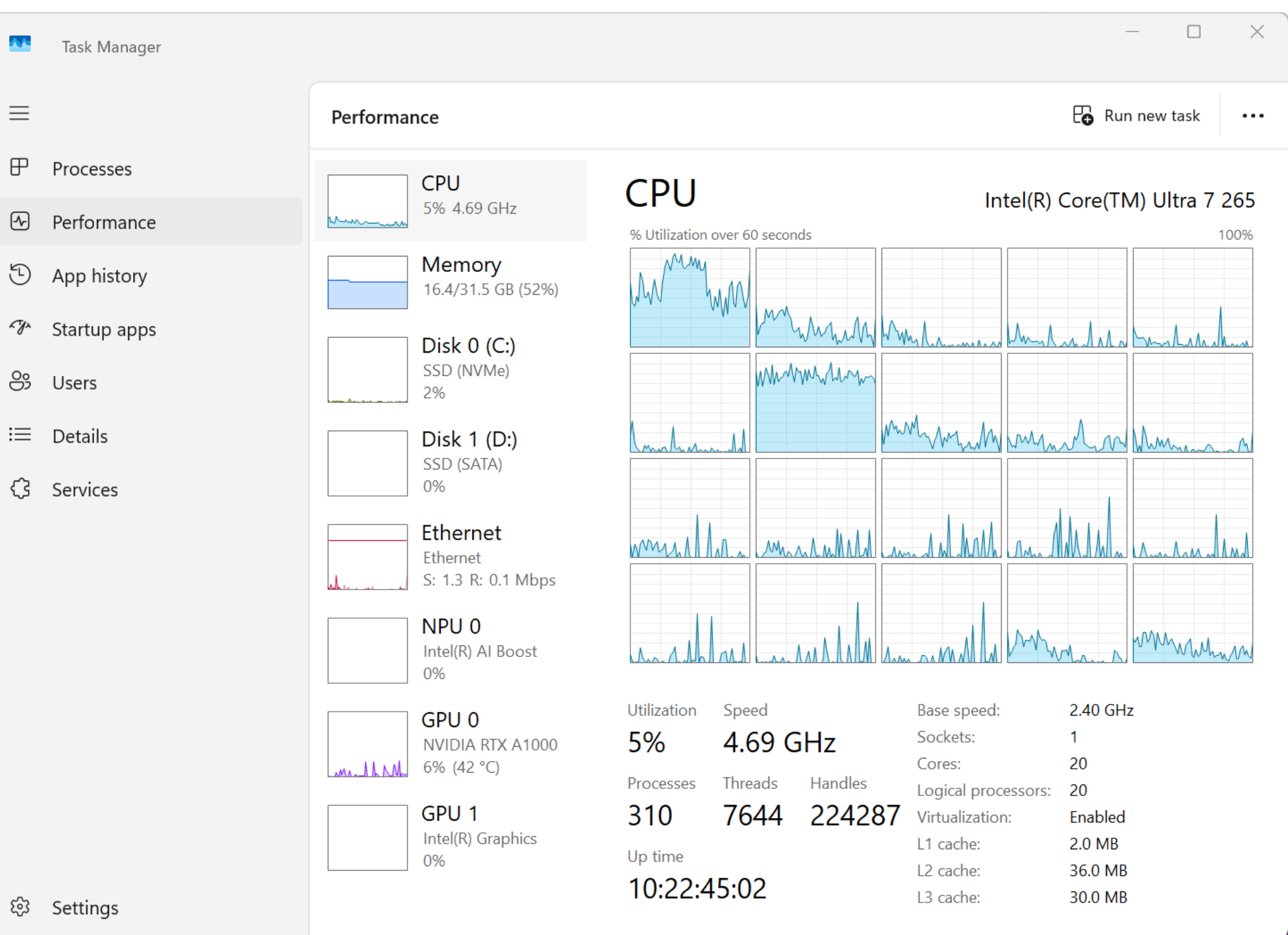
```



```
8
9 #las <- readLAS("NEON_lidar_tile.laz", filter = "-keep_random_fraction 0.5")
10 las <- readLAS("NEON_lidar_tile.laz", filter = "-thin_with_voxel 0.5")
11
12 x = pixel_metrics(las, ~length(Z)/(20*20), res=20)
13 plot(x)
```







Run singly
and check
memory

```

5 library(future)
6
7 source("Master_metrics_function.R")
8
9 # --- parallel backend ---
10 plan(multisession, workers = 3)
11
12 # --- catalog setup ---
13 ctg <- readLAScatalog('I:/neon/2019/lidar/ClassifiedPointCloud/')
14 fns <- ctg@data$filename[1:5]
15 ctg <- readLAScatalog(fns)
16 plot(ctg)
17
18 opt_chunk_alignment(ctg) <- c(0, 0)
19 opt_filter(ctg) <- '-thin_with_voxel 0.5'
20 opt_chunk_buffer(ctg) <- 20
21
22 plot(ctg, chunk_pattern = TRUE)
23
24 myfun <- function(las, res) {
25   las <- filter_poi(las, !Classification %in% c(LASNOISE, LASBUILDING))
26   las <- normalize_height(las, algorithm = tin())
27   metric_stack <- pixel_metrics(
28     las,
29     ~master_metrics(X, Y, Z, Intensity, ReturnNumber, NumberOfReturns),
30     res = res
31   )
32   return(metric_stack)
33 }
34
35 output <- lidR::catalog_map(ctg, myfun)

```


Overwrite names for memory mgmt

```
24 myfun <- function(las, res) {  
25   las      <- filter_poi(las, !Classification %in% c(LASNO  
26   las      <- normalize_height(las, algorithm = tin()))  
27   metric_stack <- nivel_metrics(/
```

plot 8		
VARIABLES		
R 4.5.1		
VALUES		
> chunk	<S4 class 'LAScluster' [package "lidR"] with 16 slots>	LAScluster
> ctg	<S4 class 'LAScatalog' [package "lidR"] with 6 slots>	LAScatalog
> ctg_full	<S4 class 'LAScatalog' [package "lidR"] with 6 slots>	LAScatalog
> fns	"I:\neon\2019\lidar\ClassifiedPointCloud\NEON_D03_JERC_DP1_731000_3459000_classified_point_clou..."	str [5]
> las	<S4 class 'LAS' [package "lidR"] with 4 slots>	LAS
> output	<S4 class 'SpatRaster' [package "terra"] with 1 slot>	SpatRaster
FUNCTIONS		
master_metrics	function (x, y, z, i, ReturnNumber, NumberOfReturns, zmin = NA, threshold = c(2, 5), dz = 1, interval_cou...	
myfun	function (las, res)	



Task Manager



Processes



Performance



App history



Startup apps



Users



Details



Services



Settings

Performance



Run new task



CPU

20% 4.69 GHz



Memory

24.7/31.5 GB (78%)



Disk 0 (C:)

SSD (NVMe)
5%



Disk 1 (D:)

SSD (SATA)
0%



Ethernet

Ethernet
S: 216 R: 152 Kbps



NPU 0

Intel(R) AI Boost
0%



GPU 0

NVIDIA RTX A1000
3% (39 °C)



GPU 1

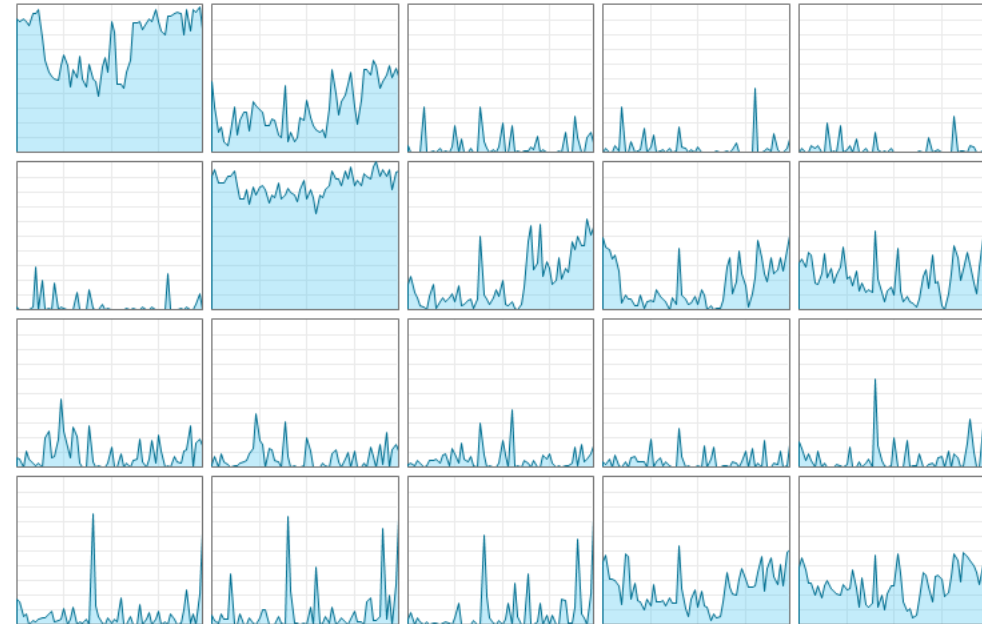
Intel(R) Graphics
0%

CPU

Intel(R) Core(TM) Ultra 7 265

% Utilization over 60 seconds

100%



Utilization

Speed

Base speed:

2.40 GHz

20%

4.69 GHz

Sockets:

1

Processes

Threads

Handles

Cores:

20

327

7800

227183

Logical processors:

20

Up time

Virtualization:

Enabled

10:23:11:27

L1 cache:

2.0 MB

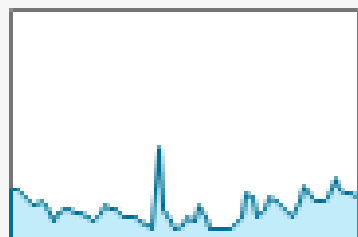
L2 cache:

36.0 MB

L3 cache:

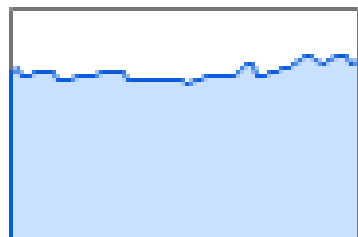
30.0 MB

Performance



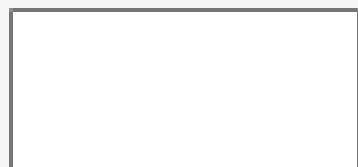
CPU

20% 4.69 GHz



Memory

24.7/31.5 GB (78%)



Disk 0 (C:)

SSD (NVMe)

Error:

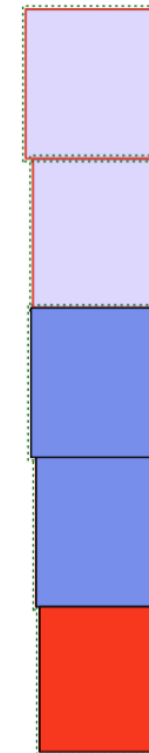
! scan() expected 'a real', got 'IllegalArgumentException:'

▼ [Hide Traceback](#) [Fix](#) [Explain](#)

1. `lidlR::catalog_map(ctg, myfun, res = 20)` at Github/codeblitz4/Master_metrics_testing_catalog.R:35:1
2. `lidlR::catalog_apply(ctg, FUN, ..., .options = .options)`
3. `lidlR::engine_apply(...)`

>

Pattern of chunks



Colors	
■	Processing
■	Empty
■	Ok
■	Warning
■	Error

```
36
37 # HOW TO TROUBLESHOOT AND RUN A SIGNLE CHUNK.
38 chunks = engine_chunks(ctg)
39 chunks[1]
40 chunk = chunks[[1]]
41 las = readLAS(chunk)
42 output = myfun(las, res=20)
43 plot(output)
44
```

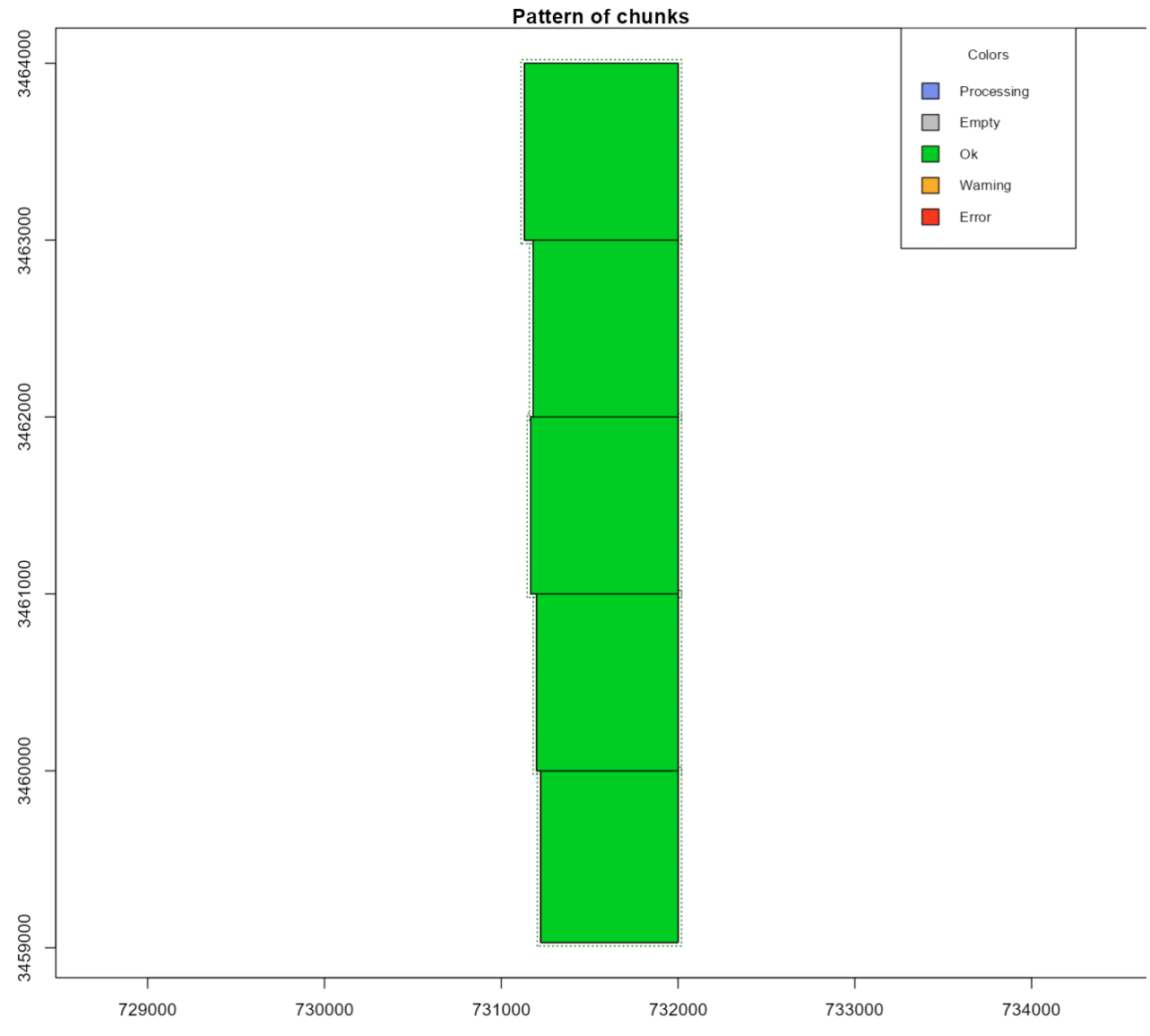
```
Master_metrics_function.R > master_metrics
```

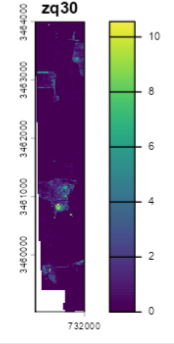
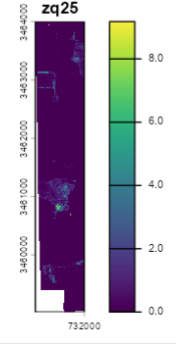
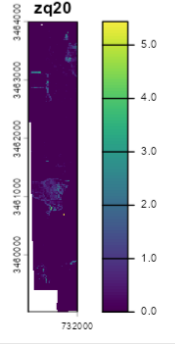
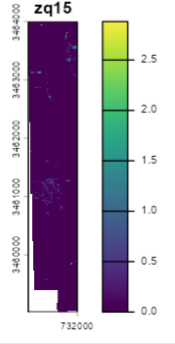
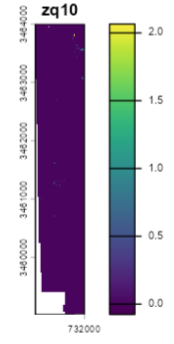
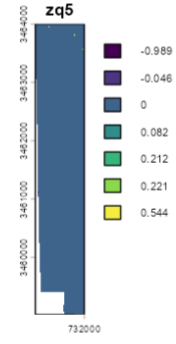
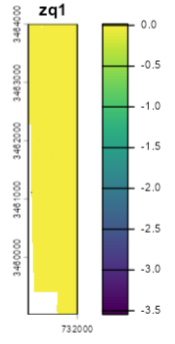
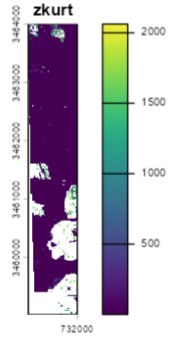
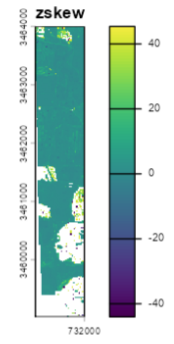
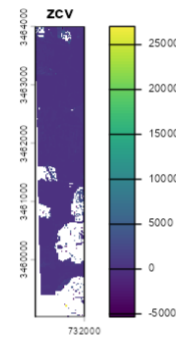
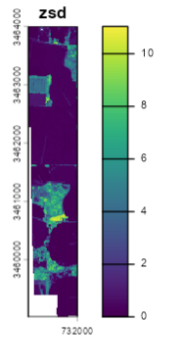
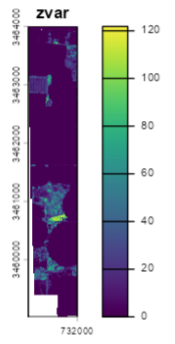
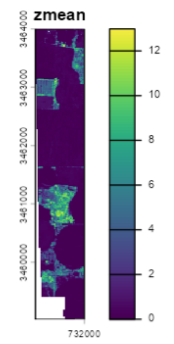
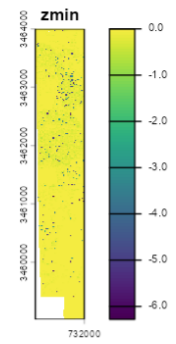
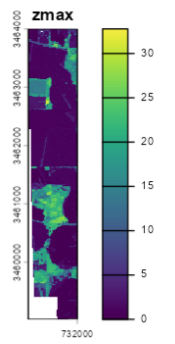
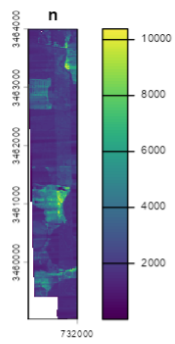
```

90 master_metrics <- function(x, y, z, i, ReturnNumber, NumberOfReturns,
117                               vox_size = vox_size, zmin = zmin)
118
119   m_kde      <- lidRmetrics::metrics_kde(z = z, zmin = zmin)
120
121   m_HOME     <- lidRmetrics::metrics_HOME(z = z, i = i, zmin = zmin)
122
123   mt         <- lidRmetrics::metrics_texture(x = x, y = y, z = z,
124                                               pixel_size = pixel_size,
125                                               zmin = zmin,
126                                               chm_algorithm = chm_algorithm)
127
128   m <- c(m_set1, m_echo, m_echo2, m_rumple, m_vox, m_kde, m_HOME, mt)
129   return(m)
130 }

```

```
90 master_metrics <- function(x, y, z, i, ReturnNumber, NumberOfReturns,  
117 |           vox_size = vox_size, zmin = zmin)  
118 |  
119   m_kde      <- lidRmetrics::metrics_kde(z = z, zmin = zmin)  
120 |  
121   m_HOME     <- lidRmetrics::metrics_HOME(z = z, i = i, zmin = zmin)  
122 |  
123+ texture_na <- list(glcm_contrast = NA, glcm_dissimilarity = NA,  
124+                   glcm_homogeneity = NA, glcm_ASM = NA,  
125+                   glcm_entropy = NA, glcm_mean = NA,  
126+                   glcm_variance = NA, glcm_correlation = NA)  
127+ |  
128+ n_cells <- nrow(unique(data.frame(floor(x / pixel_size), floor(y / pixel_size))))  
129+ |  
130+ mt <- if (n_cells >= 3) {  
131+   tryCatch(  
132+     lidRmetrics::metrics_texture(x = x, y = y, z = z, pixel_size = pixel_size,  
133+                                zmin = zmin, chm_algorithm = chm_algorithm),  
134+     error = function(e) texture_na  
135+   )  
136+ } else {  
137+   texture_na  
138+ }  
139 |  
140 m <- c(m_set1, m_echo, m_echo2, m_rumple, m_vox, m_kde, m_HOME, mt)  
141 return(m)  
142 }
```





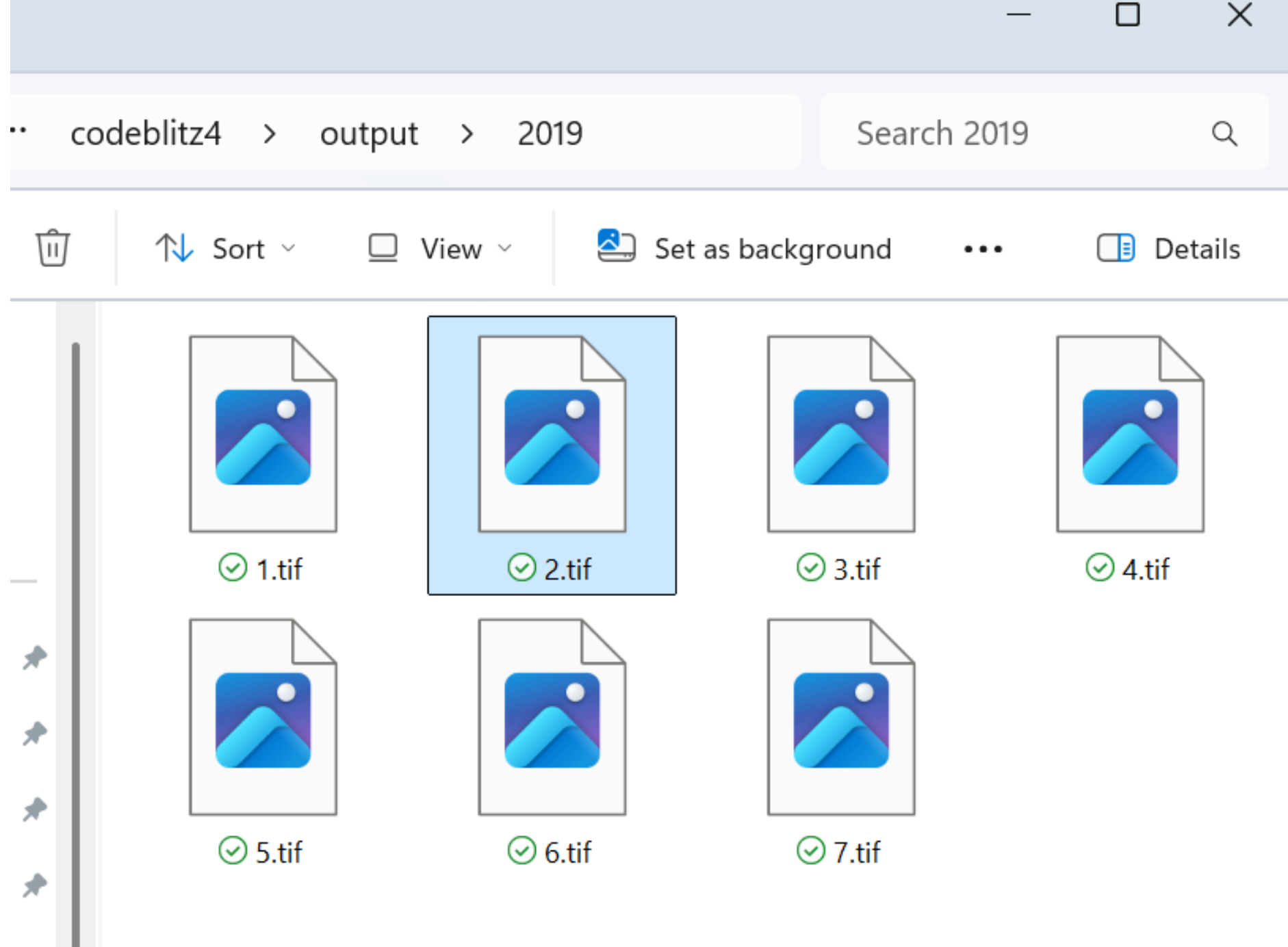
```
# =====  
# FULL CATALOG RUN – processes all tiles, saves each to disk, mosaics at end.  
# catalog_apply writes one raster per tile via opt_output_files; terra::mosaic  
# merges them into a single wall-to-wall stack.  
# =====
```

```
ctg_full <- readLAScatalog('I:/neon/2019/lidar/ClassifiedPointCloud/')  
opt_chunk_alignment(ctg_full) <- c(0, 0)  
opt_filter(ctg_full) <- '-thin_with_voxel 0.5'  
opt_chunk_buffer(ctg_full) <- 20  
opt_output_files(ctg_full) <- 'output/2019/{ID}'
```

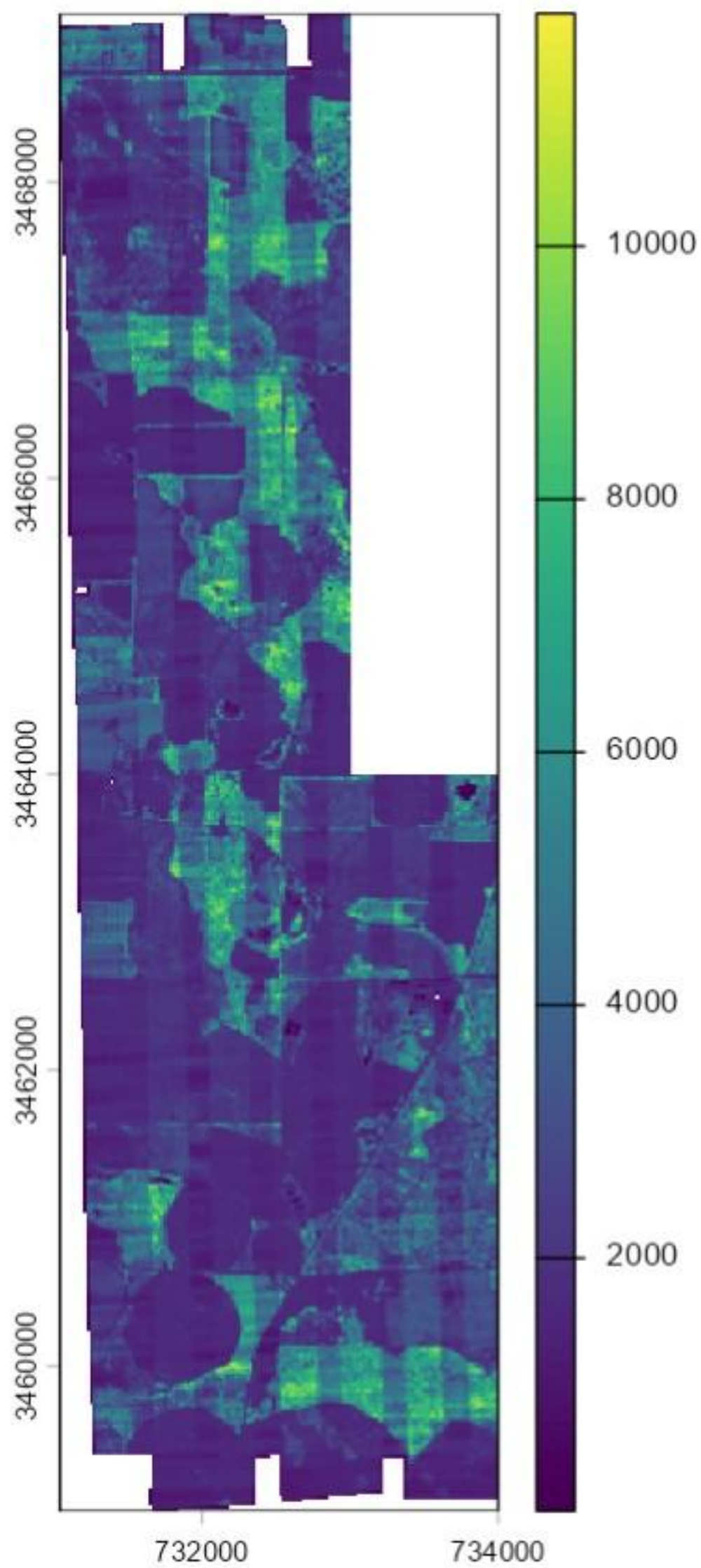
```
plan(multisession, workers = 3)  
catalog_map(ctg_full, myfun, res = 20)  
#not saving output as anything bc it will output to folder
```

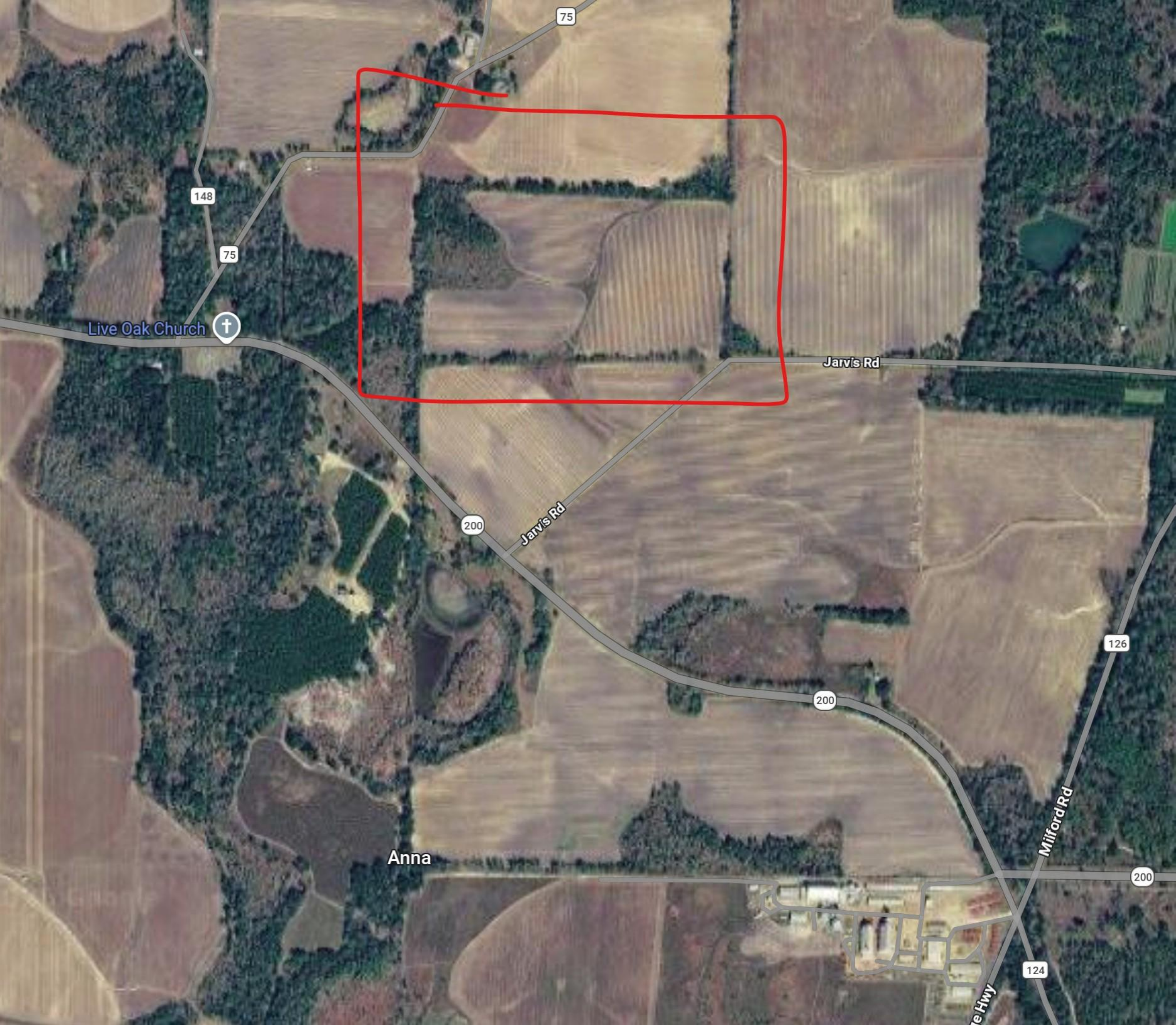
```
# mosaic all saved tiles into one raster  
tile_files <- list.files('output/2019', pattern = '\\\\.tif$', full.names = TRUE)  
tile_rasts <- lapply(tile_files, rast)  
mosaic_full <- do.call(mosaic, c(tile_rasts, list(fun = 'mean')))  
writeRaster(mosaic_full, 'output/2019_metrics.tif')
```

Save
outputs
as you go



Save
outputs
as you go





75

148

75

Live Oak Church



Jarvis Rd

200

Jarvis Rd

200

126

Anna

Milford Rd

200

124

te Hwy